Guilherme Silva Toledo

Semantics-Aware Gaussian Splatting for Egocentric Hyperlapse

Draft Version

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Advisor: Erickson Rangel do Nascimento

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Chapter 1

1.1 Tema e Justificativa

Wearable cameras such as the Project Aria glasses [1] can record hours of useful activity, but direct temporal subsampling amplifies head motion and may remove the moments that explain an event. Semantic hyperlapse addresses the second problem by assigning relevance to frames and varying the speed-up along the video [12]. Sparse semantic sampling preserves relevant content while controlling instability and output duration [16, 17]. Text-driven methods further show that relevance can be learned from language and optimized through sequential decisions [13, 14]. Emitting only recorded frames, these methods repair smoothness afterwards with stabilization and transition smoothing [15, 16]. They mainly choose among captured frames and reason mostly in image space, so they cannot place the output camera at a viewpoint that was not recorded, and semantic retention and visual smoothness remain coupled to the original shaky path.

Geometric hyperlapse follows a complementary direction. First-person Hyperlapse reconstructs the input trajectory, optimizes a smoother virtual camera path, and synthesizes its views from nearby frames [7]. Recent 3D Gaussian Splatting (3DGS) represents a scene with explicit anisotropic primitives and renders high-quality novel views in real time [6]. Lv et al. [9] adapted it to egocentric devices, modelling rolling shutter and high dynamic range on Aria recordings; its code has already been run on the entire Aria dataset for this project.

Language-embedded variants attach open-vocabulary features to this representation. LangSplat compresses CLIP features and learns object boundaries from SAM masks [11]; Semantic Gaussians projects features from several image encoders into an existing splat [4]; and EgoSplat adds multi-view instance aggregation and transient filtering for egocentric video [8]. EgoLifter similarly shows that object instances can be lifted from egocentric images into 3D [3].

Relevance can therefore be defined in 3D, on the representation that is rendered, although egocentric splats remain worse than exocentric ones [18]. No prior work plans a hyperlapse camera path over a semantic Gaussian scene; this proposal fills that gap: a semantic hyperlapse in which Gaussian-splat reconstruction provides the geometry and open-vocabulary semantics, and a path-planning stage uses that spatial-semantic model to render a smooth, fast, semantically faithful first-person video.

1.2 Objetivos

The main objective is to integrate Gaussian splatting into the semantic hyperlapse pipeline, making novel view synthesis available as in Kopf et al. [7] and decisions spatial-semantic rather than frame-wise. The specific objectives are: (i) lift open-vocabulary semantics onto the completed egocentric reconstructions, following EgoSplat [8], so that per-Gaussian relevance is queried in 3D; (ii) formulate semantic virtual camera path planning as a smooth trajectory maximizing accumulated relevance at a target speed-up, trading off proximity to the original trajectory and rendering quality; (iii) render that trajectory from the Gaussians, lifting the recorded-frame constraint that forces 2D methods to repair smoothness; and (iv) compare against both families on smoothness, semantics and motion fluidity.

1.3 Metodologia

Reconstruction and replication. The method of Lv et al. [9] has already been run on the whole Aria dataset [1], so Gaussian scenes and trajectories are available as input. We reproduce the sparse-sampling fast-forward [17] and the hyperlapse of Kopf et al. [7] on the same Aria sequences, and train LangSplat and EgoSplat features [8, 11] on the splats produced by the completed runs.

Integration. Each Gaussian receives a relevance weight from the similarity between its language feature and a set of text queries, extending the frame-level semantic score of the VeRLab pipeline to 3D. Candidate output poses are sampled around a smoothed version of the recorded trajectory. A path solver minimizes a weighted cost combining semantic visibility, deviation from the source trajectory, low render confidence, occlusion penalties, camera rotation, translational acceleration, jerk, and deviation from the requested speed-up. Selected poses are then rendered from the Gaussian scene, and frames with poor confidence fall back to nearby source views.

Evaluation. Baselines are Kopf et al. [7] and Joshi et al. [5] on the NVS side and Silva et al. [17] and Ramos et al. [14] on the semantic side, with their own metrics: semantic content retained, instability index, speed-up accuracy, segment-level F1, and PSNR, SSIM and LPIPS [19]. Data are Aria [1], Aria Digital Twin [10] and Ego-Exo4D [2].

1.4 Aspectos Éticos

The ethical concerns are limited to the data used. No new recordings of people will be made; all experiments use public egocentric datasets released under research licences with consent and anonymization already applied [1, 10], and the user study will be anonymous.

1.5 Cronograma de Execução

Reconstruction is complete and replication is under way, so the remaining work runs from January to December 2027, with the deadline at the end of 2027: baselines on the Aria sequences and first results of semantic path planning (Jan–Feb); implement semantic path planning and splat rendering, with initial experiments (Feb–May); run the experiments, ablations and user study, and write and submit the article (May–Jun); and write and defend the dissertation (Jul–Dec).

1.6 Plano de Publicação

The table below lists the target venues in date order; unannounced dates are estimated from previous editions (*est.*). SIBGRAPI 2027 is the primary target, ICCV 2027 the stretch target, and CVPR 2028 and ICLR 2028 receive the final results. ACCV has no 2027 edition and SIGGRAPH 2027 closes before the baseline is ready.

Venue	**Complete name**	**Submission**	**Conference**
ICCV 2027	*Int. Conf. on Computer Vision*	Mar 2027 (est.)	3–7 Oct 2027
SIBGRAPI 2027	*Conf. on Graphics, Patterns and Images*	Jun 2027 (est.)	Sep–Oct 2027 (est.)
ICLR 2028	*Int. Conf. on Learning Representations*	Sep 2027 (est.)	Apr–May 2028 (est.)
CVPR 2028	*Computer Vision and Pattern Recognition*	Nov 2027 (est.)	Jun 2028 (est.)

References

[1] J. Engel, K. Somasundaram, M. Goesele, et al. Project Aria: A new tool for egocentric multi-modal AI research. *arXiv preprint arXiv:2308.13561*, 2023.

[2] K. Grauman, A. Westbury, L. Torresani, K. Kitani, J. Malik, et al. Ego-Exo4D: Understanding skilled human activity from first- and third-person perspectives. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[3] Q. Gu, Z. Lv, D. Frost, S. Green, J. Straub, and C. Sweeney. EgoLifter: Open-world 3D segmentation for egocentric perception. In *European Conference on Computer Vision (ECCV)*, 2024.

[4] J. Guo, X. Ma, Y. Fan, H. Liu, and Q. Li. Semantic gaussians: Open-vocabulary scene understanding with 3D gaussian splatting. arXiv:2403.15624, 2024.

[5] N. Joshi, W. Kienzle, M. Toelle, M. Uyttendaele, and M. F. Cohen. Real-time hyperlapse creation via optimal frame selection. *ACM Transactions on Graphics*, 34(4), 2015.

[6] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis. 3D gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*, 42(4), 2023.

[7] J. Kopf, M. F. Cohen, and R. Szeliski. First-person hyper-lapse videos. *ACM Transactions on Graphics*, 33 (4), 2014.

[8] D. Li, J. Feng, J. Chen, W. Dong, G. Li, G. Shi, and L. Jiao. EgoSplat: Open-vocabulary egocentric scene understanding with language embedded 3D Gaussian splatting. *arXiv preprint arXiv:2503.11345*, 2025.

[9] Z. Lv, M. Monge, K. Chen, Y. Zhu, M. Goesele, J. Engel, Z. Dong, and R. Newcombe. Photoreal scene reconstruction from an egocentric device. In *ACM SIGGRAPH 2025 Conference Papers*, 2025.

[10] X. Pan, N. Charron, Y. Yang, S. Peters, T. Whelan, C. Kong, O. Parkhi, R. Newcombe, and Y. C. Ren. Aria digital twin: A new benchmark dataset for egocentric 3D machine perception. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

[11] M. Qin, W. Li, J. Zhou, H. Wang, and H. Pfister. LangSplat: 3D language gaussian splatting. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[12] W. L. S. Ramos, M. M. Silva, M. F. M. Campos, and E. R. Nascimento. Fast-forward video based on semantic extraction. In *IEEE International Conference on Image Processing (ICIP)*, 2016.

[13] W. L. S. Ramos, M. M. Silva, E. R. Araujo, L. S. Marcolino, and E. R. Nascimento. Straight to the point: Fast-forwarding videos via reinforcement learning using textual data. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.

[14] W. L. S. Ramos, M. M. Silva, E. R. Araujo, V. Moura, K. Oliveira, L. S. Marcolino, and E. R. Nascimento. Text-driven video acceleration: A weakly-supervised reinforcement learning method. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2022.

[15] M. M. Silva, W. L. S. Ramos, J. P. K. Ferreira, M. F. M. Campos, and E. R. Nascimento. Towards semantic fast-forward and stabilized egocentric videos. In *European Conference on Computer Vision Workshops (EPIC@ECCV)*, 2016.

[16] M. M. Silva, W. L. S. Ramos, J. P. K. Ferreira, F. C. Chamone, M. F. M. Campos, and E. R. Nascimento. A weighted sparse sampling and smoothing frame transition approach for semantic fast-forward first-person videos. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.

[17] M. M. Silva, W. L. S. Ramos, M. F. M. Campos, and E. R. Nascimento. A sparse sampling-based framework for semantic fast-forward of first-person videos. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(4), 2021.

[18] J. Warchocki, X. Wang, J. Kulhanek, and J. van Gemert. Bringing a personal point of view: Evaluating dynamic 3D Gaussian splatting for egocentric scene reconstruction. *arXiv preprint arXiv:2604.23803*, 2026.

[19] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.